Cauchy sequences of Dedekind reals converge, in the classical formulation

The Univalent Foundations Program

hott-book:reals.tex:local-snapshot-20261011; CC BY-SA 3.0 Unported

Editorial note. The counted statement is the unlabelled corollary at source lines 542-549; its original number 11.2.10 was obtained by counting the theorem-class environments of section 11.2 of the pinned source, because the environment carries no label.

Editorial note. The enumerated identity is used verbatim in the claim_key and must not be changed now that the material is registered.

Editorial note. The retained counter values (chapter 11, section 2, theorem offset 9) make the counted corollary print as Corollary 11.2.10, matching the counted original number.

Editorial note. The three external references are dump-confirmed navigation pointers invoked by the retained proof; none of them replaces a part of the counted proof, which is retained in full.

Editorial scope. One original result of the HoTT Book, counted once: convergence of Cauchy sequences of Dedekind reals (Corollary 11.2.10).

Allowed standard prerequisites (editorial).

- Cauchy condition for a sequence. A sequence x : N -> RD is Cauchy when the displayed epsilon-condition holds for all sufficiently large indices.. It is the hypothesis, referred to through the retained equation's label.
- Countable choice for properties of natural numbers. A suitable choice principle converts the pointwise Cauchy condition into a modulus.. It is the retained result invoked at source line 552 to produce M.
- Cauchy completeness of the Dedekind reals. Every Cauchy approximation of Dedekind reals has a limit.. It produces y at source lines 553-554.

Source reference. Original theorem 2.15.7: sources/hott-reals/book-labelnumbers-first-edition.tex, source lines 1909–1909. Original result cited at a distance in the chapter. It is a navigation pointer only: the counted statement's own author proof is retained line for line and nothing is replaced by it. The number is taken from the label-number dump of the pinned book snapshot; a pointer is used rather than a context span because the cited result lies in another section or is separated by unre-tained environments, so a retained context would print with the wrong number.

Source reference. Original theorem 11.2.12: sources/hott-reals/book-labelnumbers-first-edition.tex, source lines 1588–1588. Original result cited at a distance in the chapter. It is a navigation pointer only: the counted statement's own author proof is retained line for line and nothing is replaced by it. The number is taken from the label-number dump of the pinned book snapshot; a pointer is used rather than a context span because the cited result lies in another section or is separated by unre-tained environments, so a retained context would print with the wrong number.

Source reference. Original equation 11.2.9: sources/hott-reals/book-labelnumbers-first-edition.tex, source lines 1214–1214. Original result cited at a distance in the chapter. It is a navigation pointer only: the counted statement's own author proof is retained line for line and nothing is replaced by it. The number is taken from the label-number dump of the pinned book snapshot; a pointer is used rather than a context span because the cited result lies in another section or is separated by unre-tained environments, so a retained context would print with the wrong number.

Corollary 11.2.10. *Suppose* $x : \mathbb{N} \to \mathbb{R}_{\mathsf{d}}$ *satisfies the Cauchy condition* (11.2.9)*. Then there exists* $y : \mathbb{R}_{\mathsf{d}}$ *such that*

$$\prod_{(\epsilon:\mathbb{Q}_+)} \sum_{(n:\mathbb{N})} \prod_{(m\geq n)} |x_m - y| < \epsilon.$$

Proof. By original theorem 2.15.7 there is $M : \mathbb{Q}_+ \to \mathbb{N}$ such that $\bar{x}(\epsilon) :\equiv x_{M(\epsilon/2)}$ is a Cauchy approximation. Let y be its limit, which exists by original theorem 11.2.12. Given any $\epsilon : \mathbb{Q}_+$, let $n :\equiv M(\epsilon/4)$ and observe that, for any $m \geq n$,

$$|x_m - y| \leq |x_m - x_n| + |x_n - y| = |x_m - x_n| + |\bar{x}(\epsilon/2) - y| < \epsilon/4 + \epsilon/2 + \epsilon/4 = \epsilon. \ \square$$